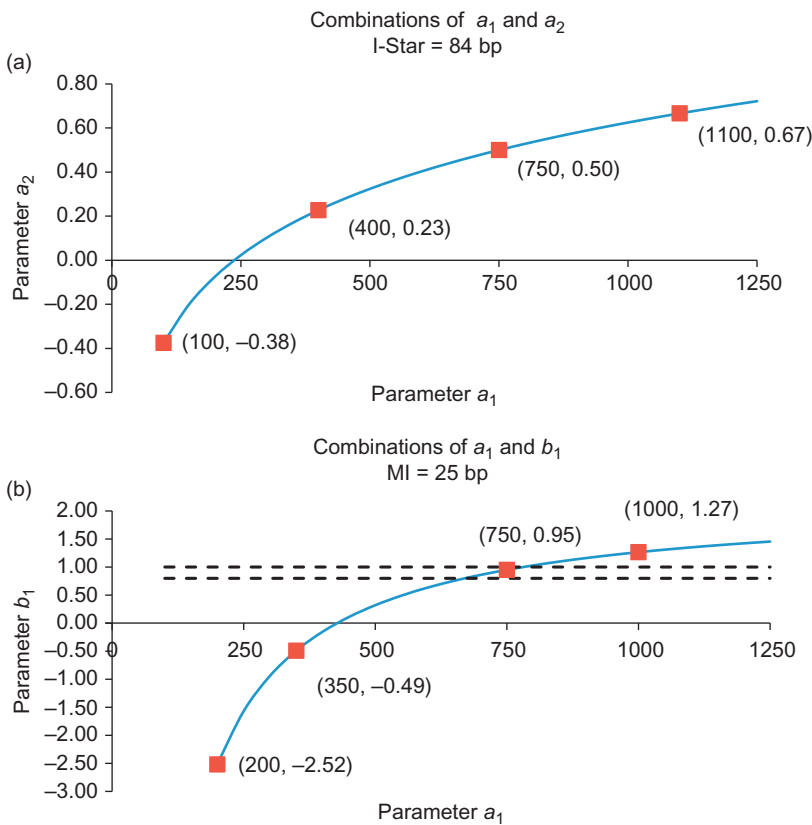


Figure 5.5a illustrates this point by showing the combinations of  $a_1$  and  $a_2$  that result in  $I^* = 84$  for  $100 < a_1 < 1250$ . But notice that for values of  $a_1 < 230$  the resulting  $a_2$  value is negative thus violating one of our feasibility requirements.

Another relationship we have with the I-Star model is with parameter  $b_1$ , which is the temporary market impact parameter—in other words, the percentage of instantaneous impact ( $I^*$ ) that is due to the liquidity needs of the investor and/or immediacy needs. Thus we have by definition  $0 \leq b_1 \leq 1$ .

Next, suppose that we have a market impact cost of  $MI = 25.3$  bp for a stock with volatility = 25% and POV rate = 30% with known parameters  $a_2 = 0.50$ ,  $a_3 = 0.75$  and  $a_4 = 0.50$ .

In this example, we have a known value of MI but the value of I-Star is not known. Then as long as the combinations of  $a_1$  and  $b_1$  result in  $MI = 84$  bp the equation is correct.



■ Figure 5.5 (a) Combinations of Parameter:  $a_1$  and  $a_2$  (b) Combinations of Parameter:  $a_1$  and  $b_1$ .